

# Umar Turdumambetov

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github.com/callmeUmar | Portfolio

## Professional Summary

Software Engineer and CS student (GPA: 4.0) building, deploying, and debugging distributed, containerized systems in C++, Python, and Java. Skilled in Linux systems programming, performance profiling, relational databases, and CI/CD automation across microservice and infrastructure tooling.

## Education

### Whitman College

Bachelor of Science in Computer Science | **GPA: 4.0**

May. 2026 – Jun. 2028

Walla Walla, WA

## Experience

### Researcher

Algoverse (RL & Post-Training Research)

Sep. 2025 – Present

Remote

- Debug and optimize distributed model-training pipelines under changing infrastructure requirements, improving stability and performance by 15–25%.
- Develop evaluation frameworks and automated test suites, reducing output variance by 20% across diverse workloads, and benchmark performance across training iterations.

### Software Engineer Intern

Canva

Nov. 2025 – Dec. 2025

Remote

- Designed and deployed modular, containerized content-generation pipelines in production, boosting throughput by 45% and reducing inconsistencies by 30%.
- Documented system architecture and collaborated cross-team on infrastructure schemas, improving overall pipeline efficiency by 35%.

### Software Engineer Intern

International University of Kyrgyzstan

Jun. 2023 – Mar. 2024

Bishkek, Kyrgyzstan

- Designed and deployed a full-stack, hardware-integrated system for 200+ users (JavaScript, FastAPI, SQL), automating 95% of a manual workflow.
- Implemented role-based access control, optimized relational database queries, and improved reliability through structured debugging and performance profiling under live production load.

## Projects

### Churn Predictor — Production System | Deployed on Render

- Designed, developed, and deployed a full-stack ML system with REST APIs and unit tests — reducing misclassification by **40%** on a **7,043-record** dataset.
- Shipped an end-to-end containerized DevOps pipeline (GitHub Actions CI/CD, Docker, automated testing) delivering releases with zero manual intervention.

### PolyPredictor-Kit — 1st Place (QuackHacks 2025) | Devpost

- Engineered a real-time, asynchronous data ingestion system with distributed FastAPI services, sustaining sub-**200ms** latency across 50+ concurrent sources.
- Structured **3,000+** raw records into production datasets, improving signal accuracy by **30%**, while optimizing concurrency and failure isolation for stable throughput.

### PromptLock — LLM Context Compression Tool | Devpost

- Designed a two-layer compression system enforcing strict **2K–16K** token budgets with rigorous unit testing across edge cases.
- Reduced failure rates by **65%** and cut inference costs up to **70%** through systematic debugging, performance optimization, and adaptive resource prioritization.

### AI Visual Novel Creator — 1st Place (CodeDay 2025) | Showcase

- Built a full-stack system with backend APIs and dynamic state routing across 20+ states under a 24-hour deadline.
- Cut processing time by **55%** and eliminated **90%+** rendering artifacts through caching and memory profiling.

## Technical Skills

**Languages:** C++, C, Python, Java, JavaScript/TypeScript (Node.js), SQL (PostgreSQL, MySQL), Bash/Shell

**Machine Learning & Data Science:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, supervised and unsupervised learning, deep learning, neural networks, NLP, computer vision, feature engineering, model evaluation, hyperparameter tuning

**Systems & Infrastructure:** Linux, distributed systems, microservices, virtualization, multi-threading, asynchronous pipelines, TCP/IP, REST APIs, gRPC

**Containers & Cloud:** Docker, Kubernetes, GitHub Actions CI/CD, AWS (EC2, S3, Lambda), Terraform

**Databases & Development Tools:** Git, GitHub, Redis, MongoDB, Apache Kafka, relational database design, query optimization

**Testing & Monitoring:** Unit testing, Postman, performance profiling, systems-level debugging, Prometheus, Grafana